

BRANDON MCCARTHY

AI Solutions & Implementation Engineer | Enterprise AI Workflows

Maryville, TN | Relocating to Cary, NC upon offer | 865-712-8581 | bmac96.dev@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

I have spent 4+ years building software through full-stack projects, applied AI systems, and technical training. I work mainly in TypeScript and Python, connecting LLMs with APIs, databases, background jobs, and AWS services. My projects use structured outputs, evaluation, human review, audit trails, and recovery paths to make AI behavior easier to test and trust. During a contract with micro1, I evaluated model responses against detailed rubrics and wrote precise feedback for model training. Five years in healthcare documentation taught me to handle complex records, protect sensitive information, and communicate clearly with clinical and legal teams.

TECHNICAL SKILLS

AI & LLM Delivery: AWS Bedrock, Claude/OpenAI APIs, RAG/pgvector, structured outputs, Zod, LLM evaluation, prompt versioning, human review

Integration & Reliability: REST APIs, webhooks, S3/SQS, Redis/BullMQ, retries/backoff, Retry-After, DLQs, idempotency, audit trails

Engineering: TypeScript, Python, SQL, React/Next.js, Node.js, FastAPI, PostgreSQL, Prisma, Drizzle, Docker

Security & Testing: multi-tenancy, RBAC/RLS, OIDC/SCIM, PHI redaction, HMAC, Playwright, Vitest, pytest, CI/CD

SELECTED PROJECTS

Enterprise AI Implementation Workbench

[Live Demo](#) | [GitHub](#)

Next.js · TypeScript · PostgreSQL/Drizzle · AWS S3/SQS/Bedrock · pgvector · Zod · Playwright

- Built a multi-tenant implementation platform that turns customer requirements and uploaded documents into cited, schema-validated delivery plans; a human approval gate controls task materialization.
- Ran AI work through S3, SQS, Bedrock, pgvector, retries/DLQ, and prompt-versioned evaluations; enforced four-role RBAC, application scoping, PostgreSQL RLS, and append-only audit history.

Pulse | Healthcare Integration Reliability

[Live Demo](#) | [GitHub](#)

Next.js · TypeScript · PostgreSQL/Prisma · Redis/BullMQ · Claude API · Docker · Playwright

- Built an investigation workspace for four simulated healthcare connectors that detects incidents from rolling health windows and produces evidence-cited findings after PHI redaction and an independent leak check.
- Gated target-revalidated recovery behind human approval; added HMAC verification, deduplication, Retry-After-aware queues, audit history, 274 automated tests, and 19 offline evaluation fixtures.

IntakeFlow | Healthcare Document Intelligence

[Live Demo](#) | [GitHub](#)

Next.js · FastAPI · PostgreSQL · OpenAI API · SQLAlchemy · Docker · Pytest

- Built a FastAPI/Next.js workflow that extracts structured fields from synthetic healthcare PDFs with page-level evidence, confidence scores, and human review for uncertain or contradictory results.
- Added deterministic validation, prompt-injection checks, idempotent processing, audit history, and a 60-packet evaluation harness; deployed the public demo on Vercel and Render.

PROFESSIONAL EXPERIENCE

micro1 | AI Trainer (Contract)

Jul 2026 - Aug 2026 | Remote

- Evaluated AI model outputs against detailed rubrics and project specifications, identifying instruction-following failures, quality gaps, and edge cases.
- Wrote precise, evidence-based feedback for model training and flagged ambiguous requirements that could affect consistent scoring.

Physician Life Care Planning | Production Specialist / Transcriptionist

Sep 2023–Present | Remote

- Prepared structured reports from complex clinical records for physicians and legal teams, averaging about 30 cases each month.
- Traced missing or conflicting source details and documented needed clarifications while following confidentiality and formatting requirements.

AQuity Solutions | Medical Scribe

Jul 2021–Sep 2023 | Remote

- Created real-time clinical documentation in EHR systems and resolved unclear details with providers.

EDUCATION & TRAINING

East Tennessee State University — B.S. Biology · Psychology minor · GPA 3.2

Aug 2014 - May 2018

App Academy — Full Stack Web Development Bootcamp

Mar 2024 - May 2025

Hack Reactor — Full Stack Web Development Bootcamp

Jun 2023 - Oct 2023

Coursera — Machine Learning Specialization (Andrew Ng)

Aug 2025